

README

USB Device Monitor

An Android app that monitors USB device attach/detach events for the **Zebra RFD40 RFID Sled** and displays live status on a modern Material UI.

Features

- **Live status indicator** — large green circle (🔌 RFD40 Awake) when the device is connected; gray circle (🔌 Sleep Mode) when disconnected
- **Device details** — shows Device Count, Vendor ID (VID 1504), Product ID (PID), and device path on connect
- **USB transport & power monitoring** — decodes the active USB mode (Power-Only / Charging vs. File Transfer / Debug) from `USB_STATE` and power connect/disconnect broadcasts
- **Interface-change counters** — running totals of USB **attach**, **detach**, and combined **total** events for the current session
- **Automatic RFID reader control** — on app startup and on RFD40 attach the app initializes the Zebra RFID SDK and connects the reader; on detach it stops using the reader (stale-session cleanup is deferred to the next attach)
- **RFD40 connection prompt** — if the reader cannot connect at startup (sled off or unplugged) the status card prompts the user to *"Turn on the RFD40 (hold its trigger to wake it) or attach it via USB"*
- **Event log** — timestamped scrollable log of all USB attach/detach, transport, power, and RFID reader events
- **Wake hint** — prompts user to *"Hold RFD40 trigger to wake up RFID Reader"* when in sleep mode
- **Toast notifications** on each attach/detach event

Screenshots

Awake (Connected)	Sleep Mode (Disconnected)
Green circle, 🔌 icon, VID/PID populated	Gray circle, 🔌 icon, wake instruction shown

Requirements

- Android **API 34+** (Android 14)
- Zebra RFD40 RFID Sled (VID 1504, PID 5889)
- USB Host mode enabled on the Android device
- **Zebra RFID API3 SDK** .aar placed in `app/libs/` (proprietary — see [app/libs/README.md](#))

Project Structure

```
app/src/main/
├─ java/com/zebra/rfid/usb/
│   ├── MainActivity.java      # USB BroadcastReceiver + UI logic
│   └─ RFIDHandler.java       # Zebra RFID SDK lifecycle (init/connect/disconnect)
├─ libs/                      # Zebra RFID API3 SDK .aar goes here
├─ res/
│   ├── layout/activity_main.xml # Material card-based UI
│   └─ drawable/
│       ├── shape_circle_awake.xml # Green status circle
│       └─ shape_circle_sleep.xml # Gray status circle
└─ build_run.sh               # Clean → Build → Install → Launch script
```

Build & Run

One-command script (requires connected device)

```
./build_run.sh
```

This script:

1. Checks for a connected ADB device/emulator
2. Runs `./gradlew clean`
3. Runs `./gradlew assembleDebug`

4. Installs the APK via `adb install -r`
5. Launches `MainActivity`
6. Streams `MYUSB` logcat output

Manual

```
./gradlew clean assembleDebug
adb install -r app/build/outputs/apk/debug/app-debug.apk
adb shell am start -n com.zebra.rfid.usb/.MainActivity
```

USB Event Handling

The app registers **two** runtime `BroadcastReceiver`s.

mUsbReceiver — host attach/detach:

Intent Action	Meaning
USB_DEVICE_ATTACHED	RFD40 plugged in — green circle, populate VID/PID
USB_DEVICE_DETACHED	RFD40 unplugged — gray circle, clear device info

mBatteryReceiver — transport mode + power state:

Intent Action	Meaning
USB_STATE	Decode transport mode via <code>updateUsbModeFromIntent()</code> (<code>connected</code> , <code>configured</code> , <code>mtp</code> , <code>ptp</code> , <code>mass_storage</code> , <code>adb</code> , <code>rndis</code>)
ACTION_POWER_CONNECTED	Cable connected — classify as <i>Power-Only (Charging)</i> or <i>File Transfer / Debug</i>
ACTION_POWER_DISCONNECTED	Cable removed — reset decoded USB state

Mode determination: explicit transport function extras (`mtp` / `ptp` / `mass_storage` / `adb` / `rndis`) take priority; if none are present, a `connected` && `configured` USB is inferred as a data-capable (file-transfer/debug) link.

See [USB_ATTACH_DETACH_DESIGN.md](#) for the full design document with code snippets.

Interface-Change Counters

The app maintains session counters for USB interface changes, shown in the **USB INTERFACE CHANGES** card:

Counter	Increments on	Color
Attach (<code>mAttachCount</code>)	<code>ACTION_USB_DEVICE_ATTACHED</code>	green
Detach (<code>mDetachCount</code>)	<code>ACTION_USB_DEVICE_DETACHED</code>	red
Total	<code>attach + detach</code> (derived)	blue

Each increment is also appended to the event log (`Attach count = N` / `Detach count = N`). Counters are session-scoped — they start at `0` on launch and are not persisted across activity recreation.

RFID SDK Integration

On startup and whenever the Zebra RFD40 sled is attached/detached, `MainActivity` drives the Zebra RFID API3 SDK through a dedicated `RFIDHandler` :

Trigger	RFIDHandler call	Action
App startup (after Bluetooth permissions)	<code>startup()</code>	Initialize SDK → enumerate readers → connect; on failure show the RFD40 prompt
USB_DEVICE_ATTACHED (VID 1504)	<code>onUsbAttached()</code>	Clean up any stale session → initialize SDK → enumerate readers → connect → configure event listeners
USB_DEVICE_DETACHED	<code>onUsbDetached()</code>	Best-effort stop the reader's read pipeline (so the SDK IO manager stops polling the dead endpoint); full SDK cleanup is deferred to the next attach
<code>onDestroy()</code>	<code>onDestroy()</code>	Dispose SDK + shut down background executor

The Zebra RFID SDK enumerates bonded Bluetooth devices on init, so on API 31+ `MainActivity` requests the runtime `BLUETOOTH_CONNECT` / `BLUETOOTH_SCAN` permissions on launch and runs the startup connect once they are granted.

Reader status (connecting, connected, disconnected, tag reads, errors) is routed back into the event log with a `[RFID]` prefix via a `StatusListener` callback. The same listener also reports a terminal connection result — `onRfidConnectionResult(connection, message)` — that drives the status card: a green **RFD40 Connected** state on success, or an amber **RFD40 Not Connected** prompt on failure. SDK work runs on a single-threaded background executor with volatile guards (`initializationInProgress`, `connectionInProgress`, `resumeRequested`, `readersAttached`) to avoid duplicate init/connect races.

SDK requirement: the proprietary Zebra RFID API3 `.aar` is **not** bundled in this repository. Drop it into `app/libs/` (see [app/libs/README.md](#)). `app/build.gradle` pulls it in via `implementation fileTree(dir: 'libs', include: ['*.aar', '*.jar'])`. Until present, the `com.zebra.rfid.api3.*` symbols in `RFIDHandler.java` will not resolve and the project will not build.

See [USB_ATTACH_DETACH_DESIGN.md](#) §10 for the full integration design.

Key Log Tags

```
adb logcat -s MYUSB:D MYUSB_STATE:D
```

```
D Device List Size: 1
D Device VID: 1504
D Device PID: 5889
D USB action: android.hardware.usb.action.USB_DEVICE_DETACHED
D USB device detached
D USB action: android.hardware.usb.action.USB_DEVICE_ATTACHED
D ACTION_USB_DEVICE_ATTACHED
D --> Vid =1504
```

USB transport/power events are logged under the `MYUSB_STATE` tag:

```
D USB_STATE[broadcast] connected=true, configured=true, mtp=false, ptp=false, mass_storage=false, adb=true, rndis=false, explicitDataModeKnown=t
D ACTION_POWER_CONNECTED mode=File Transfer / Debug, connected=true, configured=true, fileTransfer=true
D ACTION_POWER_DISCONNECTED
```

License

Apache 2.0